

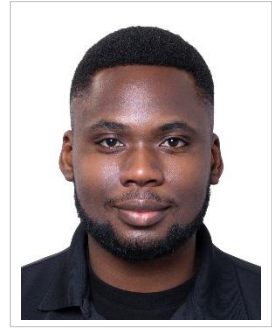
CHRISTOPHER ASH

AI Engineer

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Portfolio: chrisdanash.github.io/ChrisDanAsh/Portfolio/CAsh_EN.pdf

GitHub: github.com/ChrisDanAsh



PROFESSIONAL SUMMARY

AI engineer whose work spans agentic LLM systems, applied ML and embedded systems, most recently shipping a multilingual planner-tool assistant to production and raising task completion from 55% to 84%. A Global Korea Scholarship recipient who is curious, quick to adapt to any challenge, and holds integrity and ethics above all else.

TECHNICAL SKILLS

Programming & Frameworks: C, C++, Python, PyTorch, SQL, LangChain, GCP, Docker

AI & Research: LLM prompt engineering, Retrieval-Augmented Generation (RAG) systems, Agentic LLM systems, Time series analysis & prediction, transfer learning

Embedded Hardware: Arduino, Raspberry Pi, PIC microcontrollers, motor controllers, sensor/actuator systems

Languages: English (native), Korean (TOPIK Level 4 – Intermediate)

Other: Team leadership

WORK EXPERIENCE

AI Engineer

Jan 2026 – June 2026

TripDoc, Korean Medical Tourism Company

- Responsible for the entire process of design to production of an agentic multilingual LLM based chat capable of web search backed health advice, finding hospitals, treatments and pharmacies, as well as making treatment reservations using natural language
- Built a FastAPI-based REST and SSE streaming API backend: Asynchronous SQLAlchemy persistence, session state management and caching, concurrency control, timeouts, Docker containerization, and GCP environment operations
- Designed and executed a 25-scenario multilingual benchmark comparing it to Gemini and ChatGPT; diagnosed structural errors and drove improvements to improve the AI chat completion rate from 55% to 84%
- Transformed benchmark results into prioritized, step-by-step engineering roadmap reports and documented the end-to-end system architecture
- Improved a medical domain-specific LLM translation API: DB-based term masking, improved prompts, and separated endpoints for internal translation and external live chat

Student Researcher

Sept 2023 – May 2025

Artificial Intelligence Application Lab, Kyungpook National University

- Conducted research in time series analysis and prediction using AI (Python / PyTorch)
- Authored peer review reports for AUTCON and EAAI submissions under supervision, evaluating methodology, results, and clarity
- Translated lab research papers and documentation from Korean to English for publication and internal review

R&D Engineer

Sept 2020 – Nov 2021

The University of the West Indies, Trinidad

- Responsible for research, design and development of an Autonomous Disinfection Robot and an Air Disinfection Unit
- Implemented the motor controller for the Disinfection Robot on an Arduino
- Implemented the sensor and actuator system for the Air Disinfection Unit on a Raspberry Pi

Research Assistant

Nov 2019 – Oct 2020

The University of the West Indies, Trinidad

- Screened and synthesised the state of the field for a real-time fault-tolerant control review paper
- Ran tutorial classes for the Controls and Instrumentation I course

EDUCATION

MSc. Robotics and Smart Systems Engineering, AI and Safety Convergence (Double Major)*Kyungpook National University*

Sept 2023 – Aug 2025

GPA: 3.9 / 4.3

Korean Language Program (Global Korea Scholarship)*Dongseo University*

Sept 2022 – Aug 2023

BSc. Electrical and Computer Engineering*The University of the West Indies, Trinidad*

Sept 2015 – July 2019

First Class Honours • GPA: 3.73 / 4.3

PUBLICATIONS

Air Pollution Spread Prediction and Risk Assessment – A Sequential Transfer Learning Model for Air Pollution Prediction in Data Scarce Regions. *KICEM Annual Conference, 2024*

ACADEMIC ACHIEVEMENTS

2022	Global Korea Scholarship Winner, Trinidad and Tobago
2019	Winner of iGovTT HACKTT App Development Competition – Group Project Manager
2019	Faculty of Engineering Dean's Honour Roll, University of the West Indies St. Augustine
2018	Board of Engineering of Trinidad and Tobago Bursary recipient
2015	Trinidad and Tobago National Scholarship – Natural Sciences
2014, 2015	Top CXC CAPE Unit I and II Student (Male), St. Stephen's College, Trinidad

VOLUNTEER EXPERIENCE & ACTIVITIES

2024 – 2025	KNU International Exchange Club Member
2018 – 2019	Engineering Student Society Secretary, UWI St. Augustine
2015 – 2019	Student Liaison for Electrical and Computer Engineering Department
2017 – 2019	Instructor at Engineering Programming Workshops (Jan 2019, 2018, 2017)
2013 – 2015	Chess Club President, St. Stephen's College